



Minerals (Nutrition and measurements)

Minerals : are the chemical elements present in all body tissues and fluids , their presence being necessary for the maintenance of certain physicochemical processes which are essential for life.

Minerals Nutrients: are inorganic substances that must be ingested and absorbed in adequate amounts to satisfy a wide variety of essential metabolic and/or structural functions in the body.

Minerals and electrolyte are important **because** your body cannot function without them. All electrolyte are minerals but not all minerals are electrolyte .Electrolyte are the minerals that carry an electrical charge.

Your body requires some raw materials. These include at least 30 vitamins, minerals, and dietary components that your body needs but cannot manufacture on its own in sufficient amounts.

Vitamins and minerals are considered essential nutrients—because acting in concert, they perform hundreds of roles in the body. They help shore up bones, heal wounds, and bolster your immune system. They also convert food into energy, and repair cellular damage.

They are classified as major minerals(macrominerals) (body requires more than 100 mg/day) and trace minerals (microminerals)(body requires less than 100 mg/day), and The two tables below list minerals, what they do in the body (their functions), and their sources in food.



Major minerals

Mineral	Function	Sources
Sodium	Needed for proper fluid balance, nerve transmission, and muscle contraction	Table salt, soy sauce; small amounts in milk, breads, vegetables.
Chloride	Needed for proper fluid balance, stomach acid, maintain the osmosis pressure in body cell.	Table salt, soy sauce; small amounts in milk, meats, breads, and vegetables
Potassium	Needed for proper fluid balance, nerve transmission, and muscle contraction	Meats, milk, fresh fruits and vegetables, whole grains, legumes
Calcium	Important for healthy bones and teeth; helps muscles relax and contract; important in nerve functioning, blood clotting, blood pressure regulation, immune system health.	Milk and milk products; canned fish with bones (salmon, sardines); greens (broccoli), legumes.
Phosphorus	Important for healthy bones and teeth; found in every cell; part of the system that maintains acid-base balance.	Meat, fish, poultry, eggs, milk, processed foods.
Magnesium	Found in bones; needed for making protein, muscle contraction, nerve transmission, immune system health.	Nuts and seeds; legumes; leafy, green vegetables; seafood; chocolate.

- ❖ (calcium ,magnesium and phosphor are bulk mineral, then (sodium ,potassium and chloride) are electrolytes.
- ❖ Trace minerals (microminerals) The body needs trace minerals in very small amounts.



Trace minerals		
Mineral	Function	Sources
Iron	Part of a molecule (hemoglobin) found in red blood cells that carries oxygen in the body; needed for energy metabolism	Organ meats; red meats; fish; poultry; shellfish ,egg yolks; legumes; dried fruits.
Zinc	Part of many enzymes; needed for making protein and genetic material; has a function in taste perception, wound healing, normal fetal development, immune system health	Meats, fish, poultry, leavened whole grains, vegetables
Iodine	Found in thyroid hormone, which helps regulate growth, development, and metabolism	Seafood, foods grown in iodine-rich soil, iodized salt, bread.
Selenium	Antioxidant	Meats, seafood, grains
Copper	Part of many enzymes; needed for iron metabolism	Legumes, nuts and seeds, organ meats, drinking water
Fluoride	Involved in formation of bones and teeth; helps prevent tooth decay	Drinking water (either fluoridated or naturally containing fluoride), fish
Chromium	Works closely with insulin to regulate blood sugar (glucose) levels	Unrefined foods, especially liver, whole grains, nuts, cheeses



Daily Requirements

The daily requirements of minerals required by the body can be obtained from a well balanced diet. Like vitamins, excess minerals can produce toxic effects.

The following table should cover the recommended daily requirements (RDR or RDA) of almost.

Mineral	Men	Women
Calcium	1000mg	1200mg
Sodium	1100mg	3300mg
Potassium	2000mg	2000mg
Iron	10mg	15mg
Zinc	15mg	12mg
Magnesium	350mg	280mg
Phosphorus	800mg	1200mg
Chlorine	700mg	700mg
Fluorine	1.5mg	4mg
Copper	2mg	2mg
Selenium	0.07mg	0.05mg

Excess of minerals : High concentrations of minerals in the body can have ill effects on your health. Symptoms of mineral toxicity vary as toxic levels depend on the type of mineral and how much of it your body takes in.

High levels of iodine in the bloodstream can interfere with hormone functioning. Too much sodium can cause confusion, seizures coma and even death. Selenium is a mineral that is toxic in just small doses. Symptoms include black fingernails and the smell of garlic on your breath and skin. Phosphorus toxicity prevents the absorption of calcium and magnesium in the body. When ingested in amounts more than **1g** daily, phosphorus can cause diarrhea or lead to calcification of organs and soft tissues.



Copper toxicity is also rare, however, excessive intake can cause vomiting, diarrhea, irritability and dementia.

Zinc is generally considered to be non-toxic although extremely high doses of it can lead to symptoms such as nausea vomiting and diarrhea.

Mineral Measurement :

Measurement of minerals is a commonly performed diagnostic procedure, performed via blood testing

1. Flame photometer or (Atomic absorption spectroscopy AAS):

A device used in organic chemical analysis to determine the concentration of minerals ions, such as (Ca, k and Na) .

2. Auto analyzer (biochemistry analyzer):

An instrument with ion selective electrodes ,which measured electrolyte most often are sodium and potassium, chloride and bicarbonate or total CO₂.

3. Dual-energy X-ray absorptiometry (DXA):

used to measure total body composition. The DXA scan is typically used to diagnose and follow osteoporosis, The bone density test is painless and quick. The X- rays measure how much calcium and minerals are in a part of your bone.



The amount of phosphate in the blood affects the level of calcium in the blood. Calcium and phosphate in the body react in opposite ways: as blood calcium levels rise, phosphate levels fall. A hormone called parathyroid hormone (PTH) regulates the levels of calcium and phosphorus in your blood. When the phosphorus level is measured, a vitamin D level, and sometimes a PTH level, is measured at the same time. Vitamin D is needed for your body to take in phosphate.

The relation between calcium and phosphate may be disrupted by some diseases or infections. For this reason, phosphate and calcium levels are usually measured at the same time.